**R15/R13**

**Code No: 125EH/115EH**

**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD**
**B. Tech III Year I Semester Examinations, March - 2021**
**OPERATING SYSTEMS**
**(R15-Common to CSE, IT; R13-Common to CSE, IT)**

**Time: 3 hours** **Max. Marks: 75**

**Answer any five questions**
**All questions carry equal marks**
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1.a) What systems calls have to be executed by a command interpreter or shell in order to start a new process?
b) Why do some operating systems store the operating system in firmware, while others store it on disk. [7+8]

2.a) Distinguish between the client server and peer to peer models of distributed systems.
b) Describe the three general methods for passing parameters to the operating system. [7+8]

3. What are the completion times of processes P1, P2 and P3 using SRTF (Shortest Remaining Time First)? [15]

| | **PID** | **AT** | **BT** | |
|---|---|---|---|---|
| | CPU | IO | CPU | |
| P1 | 0 | 1 | 2 | 2 |
| P2 | 1 | 2 | 4 | 5 |
| P3 | 2 | 3 | 6 | 8 |

4.a) Which critical section problem satisfies all three conditions of critical section problem to two processes that alter execution between their critical section and remainder sections?
b) Draw and explain process state diagram with two suspend states. [8+7]

5. Let the average process size be s bytes and the page size is p bytes. Furthermore, assume that each page entry requires e bytes. Derive the optimal page size. [15]

6.a) Define Segmentation. Explain the Segmentation with Paging.
b) Derive a formula for optimal page size. [9+6]

7. Explain different disk allocation methods in detail. [15]

8. Discuss the strengths and weakness of implementing an access matrix using access list that are associated with domains. [15]

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USED PAPER MARCH-2021